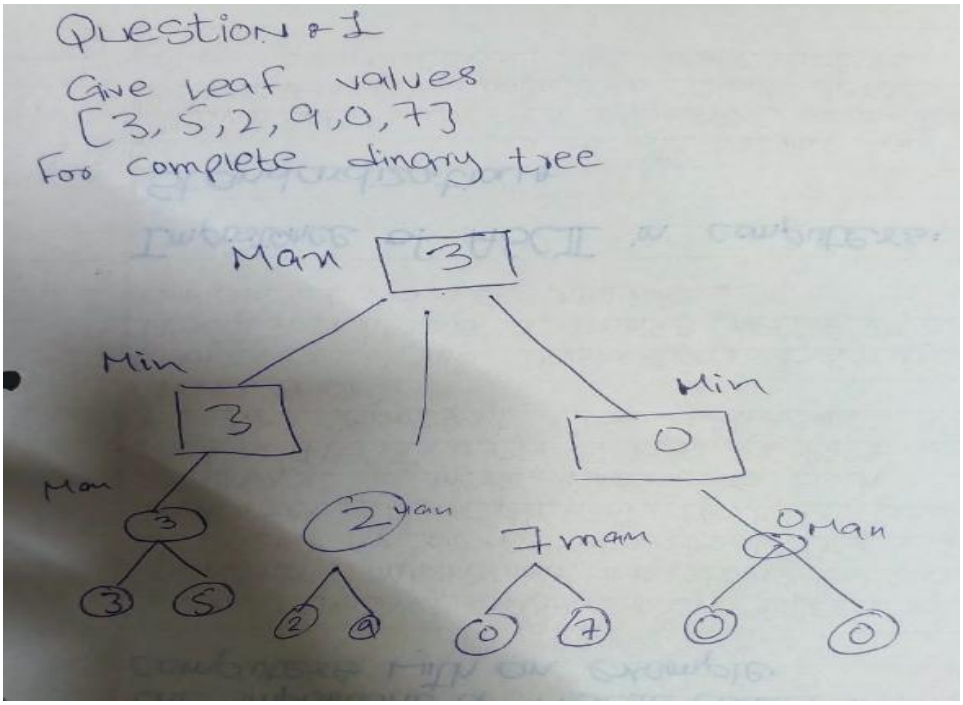
NAME: AZEEM **ROLL NO: 039**

CLASS: BSAI **SECTION: 3A**

LAB TASK NO: 5

Question:1) Minimax Hand Trace



Question:2) Leaf Evaluation Counter

Minimax checks all 8 leaves.

Alpha-Beta skips unnecessary leaves.

```
def minimax(scores, depth, index, max_turn, count):
    if depth == 3:
        count[0] += 1
        return scores[index]
if max_turn:
        return max(
      minimax(scores, depth+1, index*2, False, count),
           minimax(scores, depth+1, index*2+1, False, count))
```

```
    else:
        return min(
            minimax(scores, depth+1, index*2, True, count),
            minimax(scores, depth+1, index*2+1, True, count))
def alphabeta(scores, depth, index, max_turn,
              alpha, beta, count):
if depth == 3:
        count[0] += 1
        return scores[index]
if max_turn:
        value = float('-inf')
for i in range(2):
            value = max(value, alphabeta(
                scores, depth+1, index*2+i,
                False, alpha, beta, count))
            alpha = max(alpha, value)
if alpha >= beta:
                break
return value
else:
  value = float('inf')
      for i in range(2):
            value = min(value, alphabeta(
                scores, depth+1, index*2+i,
                True, alpha, beta, count))
            beta = min(beta, value)
if alpha >= beta:
                break
   return value
scores = [3, 5, 2, 9, 0, 7, 0, 0]
```

```
m_count = [0]
a_count = [0]
m = minimax(scores, 0, 0, True, m_count)
a = alphabeta(
    scores, 0, 0, True,
    float('-inf'), float('inf'), a_count)
print("Minimax:", m)
print("Minimax Leaves:", m_count[0])
print("Alpha-Beta:", a)
print("Alpha-Beta Leaves:", a_count[0])
new_scores = [3, 5, 2, 9, 0, 0, 0, 7]
new_count = [0]
new_value = alphabeta(
    new_scores, 0, 0, True,
    float('-inf'), float('inf'), new_count)
print("Reordered Value:", new_value)
print("Reordered Leaves:", new_count[0])
```

Question:3) Heuristic Overestimate

Original:

[3, 5, 2, 9, 0, 7, 0, 0]

Reordered:

[3, 5, 2, 9, 0, 0, 0, 7]

Results

Test	Minimax value	Leaves evaluated
Minimax	5	8
Alpha-Beta original	5	8
Alpha-Beta reordered	5	6

LAB TASK 5 REPORT

Objective

Minimax was traced by hand and compared with Alpha-Beta pruning in Python on a binary tree with 8 leaves, counting how many leaves each evaluates.

Results

Test	Value	Leaves evaluated
Minimax	5	8
Alpha-Beta (original order)	5	8
Alpha-Beta (reordered)	5	6

Conclusion

Both algorithms return 5. Alpha-Beta still checked all 8 leaves on the original order, but only 6 after swapping 7 and 0, because the leaves (0, 7) were pruned once the MIN node fell below the root's alpha of 5. Move ordering decides how much Alpha-Beta saves.